

Option Pricing Calculations

This document describes the mathematical models and formulas used to price options in the system. Each asset class is presented on its own page with clean, presentation-ready notation.

Equity Options – Black-Scholes Model

Equity options are priced using the Black-Scholes framework. The underlying asset is the spot price of a stock.

Model Inputs:

- S : Stock spot price
- K : Strike price
- σ : Annualised volatility
- r : Continuously compounded risk-free interest rate
- T : Time to maturity in years

Intermediate Calculations:

- $d1 = (\ln(S / K) + (r + 0.5 * \sigma^2) * T) / (\sigma * \sqrt{T})$
- $d2 = d1 - \sigma * \sqrt{T}$
- Discount Factor = $\exp(-r * T)$

Option Prices:

- Call Price = $S * N(d1) - K * \exp(-r*T) * N(d2)$
- Put Price = $K * \exp(-r*T) * N(-d2) - S * N(-d1)$

Primary Greeks:

- Delta : Sensitivity to stock price movements
- Gamma : Curvature of delta
- Vega : Sensitivity to volatility
- Theta : Time decay of option value
- Rho : Sensitivity to interest rates

Bond Futures Options – Black-76 Model

Bond futures options are priced using the Black-76 model. The underlying asset is the futures price, not the bond spot price.

Model Inputs:

- F_0 : Current bond futures price
- K : Strike price
- σ : Volatility of the futures price
- r : Risk-free interest rate
- T : Time to maturity in years
- Contract Size : Futures notional amount

Intermediate Calculations:

- $d_1 = (\ln(F_0 / K) + 0.5 * \sigma^2 * T) / (\sigma * \sqrt{T})$
- $d_2 = d_1 - \sigma * \sqrt{T}$
- Discount Factor = $\exp(-r * T)$

Option Prices:

- Call Price = $\exp(-r*T) * (F_0 * N(d_1) - K * N(d_2))$
- Put Price = $\exp(-r*T) * (K * N(-d_2) - F_0 * N(-d_1))$

Final option value is obtained by multiplying the unit price by the contract size.

SOFR Futures Options – Black-76 Model

SOFR options are options on SOFR futures contracts. The futures price is quoted as 100 minus the implied SOFR rate.

Model Inputs:

- F_0 : SOFR futures price (for example, 95.50 implies a 4.50 percent rate)
- K : Strike price
- σ : Volatility of the futures price
- r : Risk-free interest rate
- T : Time to maturity in years
- Contract Size : Typically around 2,500,000

Intermediate Calculations:

- $d_1 = (\ln(F_0 / K) + 0.5 * \sigma^2 * T) / (\sigma * \sqrt{T})$
- $d_2 = d_1 - \sigma * \sqrt{T}$
- Discount Factor = $\exp(-r * T)$

Option Prices (per futures unit):

- Call Price = $\exp(-r*T) * (F_0 * N(d_1) - K * N(d_2))$
- Put Price = $\exp(-r*T) * (K * N(-d_2) - F_0 * N(-d_1))$

Scaling and Interest Rate Sensitivity:

- Final Option Value = Unit Price * Contract Size
- Rho (approximation used in demo) = $-T * \text{Option Price}$